

Q. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible Cause	How to Diagnose	How to Fix
1. Wi-Fi is turned off or not connected	Check if the Wi-Fi icon shows "Disconnected" or if Airplane Mode is enabled.	Turn on Wi-Fi, disable Airplane Mode, and reconnect to the correct Wi-Fi network (SSID). Enter the correct Wi-Fi password if required.
2. Laptop did not receive an IP address	Open Command Prompt and type ipconfig . If the IP address starts with 169.254.x.x , the laptop did not get an IP address from the router.	Restart the router, then type ipconfig /release followed by ipconfig /renew to request a new IP address.
3. Router or Internet connection is down	Check whether other devices (such as the second laptop or Smart TV) can access the internet. Also check if the router's Internet light is on.	Restart the router and modem. If no devices have internet, contact the Internet Service Provider (ISP).

Additional Troubleshooting Steps

1. Restart the laptop and the Wi-Fi router.
2. Make sure the laptop is connected to the correct Wi-Fi network.
3. Check if the Wi-Fi password is correct.
4. Run the Windows **Network Trouble-shooter**.
5. Open Command Prompt and test the connection:
 - o ping 192.168.1.1 → Tests the connection to the router.
 - o ping 8.8.8.8 → Tests whether the internet is reachable.
 - o ping google.com → Checks whether DNS is working.